

OCR Engine Resource Analysis Report

System Resource Classification for Tier Assignment
Unified OCR Platform Engineering Team | June 24, 2026

Executive Summary

This report categorizes OCR engines by system resource requirements to inform tier-based pricing and infrastructure allocation. Analysis covers four model families deployed across Free, Basic, Pro, and Enterprise tiers.

Key Finding: Resource footprint ranges from 0.9B to 4B parameters, with compute cost varying by 85% between lightest and heaviest models.

1. Resource Classification

1.1 Primary Engines

Engine	Parameters	GPU Requirement	VRAM	Resource Class
PaddleOCR Lite	-	Optional	<2 GB	Light (CPU-viable)

Engine	Parameters	GPU Requirement	VRAM	Resource Class
PaddleOCR-VL	0.9B	CUDA	3-4 GB	Light (compact)
GOT-OCR 2.0	580M	CUDA	4-6 GB	Light-Medium
Infinity-Parser2-Flash	2B	Auto	5-7 GB	Medium
Qianfan-OCR	4B	Auto	10-14 GB	Heavy

Table 1: Resource requirements by OCR engine

1.2 Cost Profile Mapping

Engine	Cost Profile	Current Tier	Rationale
PaddleOCR Lite	Low	Free	Entry-level, minimal resources
PaddleOCR-VL	Low	Basic	Compact, high efficiency
Infinity-Parser2-Flash	Low	Pro	Flash variant, throughput optimized
GOT-OCR 2.0	Medium	Pro	Balanced performance/cost
Qianfan-OCR	Medium	Pro	Advanced capabilities
GOT-OCR 2.0 Enterprise	High	Enterprise	Full feature set

Engine	Cost Profile	Current Tier	Rationale
Qianfan-OCR Enterprise	High	Enterprise	Maximum capability
Infinity-Parser2 Ent.	Medium	Enterprise	Extended features

Table 2: Cost profiles and tier assignments from platform configuration

2. Tier Recommendations

2.1 Resource-Based Tier Structure

Tier	Resource Budget	Assigned Engines
Free	<2 GB VRAM, CPU-viable	PaddleOCR Lite
Basic	3-4 GB VRAM, T4 GPU	PaddleOCR-VL
Pro	5-7 GB VRAM, T4/V100	GOT-OCR 2.0, Qianfan-OCR, Infinity-Flash
Enterprise	10-14 GB VRAM, A100	All engines + extended quotas

Table 3: Infrastructure requirements by tier

2.2 Key Insights

- **Light engines (Free/Basic):** PaddleOCR family supports high-volume, cost-sensitive workloads. PaddleOCR-VL achieves SOTA table accuracy at 0.9B params.

- **Medium engines (Pro):** Three specialized engines cover different use cases:
 - GOT-OCR: Academic papers, formatted output (580M, smallest neural)
 - Infinity-Flash: Fast document parsing, 3.68× throughput vs 7B predecessor
 - Qianfan: Complex layouts, KIE (4B, highest intelligence)
- **Computational spread:** 85% cost difference between GOT-OCR (580M) and Qianfan (4B) enables precise workload-to-resource matching.

3. Technical Specifications

3.1 Platform Configuration

Precision BF16 (except GOT-OCR)	PDF DPI 144 (configurable)
Max Pages 50 per document	Memory Management Auto-unload enabled

3.2 Scaling Characteristics

- **Cold start:** 15-45 seconds (mitigated via EAGER_LOAD in production)
- **Throughput:** Inversely correlated with model size (Infinity-Flash highest)
- **Concurrent requests:** Smaller models (GOT-OCR) support more parallel instances per GPU

4. Recommendations

1. Maintain current tier mapping: Resource analysis validates existing assignments.

2. Infrastructure provisioning:

- Free/Basic: CPU or T4 instances
- Pro: V100 instances (cost-optimal for 3 concurrent engines)
- Enterprise: A100 for Qianfan-OCR heavy workloads

3. Enable eager loading: Set `EAGER_LOAD=true` for all Pro/Enterprise engines in production to eliminate cold start latency.

4. Benchmark priority: Measure actual latency on target GPU infrastructure for cost modeling refinement.

Data Sources: Platform configuration (seed.py, config.py), knowledge base, official model papers

This report supports infrastructure planning and tier-based resource allocation for the Unified OCR Platform.